

Brian Kominick

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SUMMARY

Software engineer with 5+ years of experience building automation frameworks, backend services, and developer tooling. Currently pursuing a thesis-based M.S. in computer science, with research focused on applied generative AI and performance-aware systems at the compiler and infrastructure level.

ENGINEERING HIGHLIGHTS

- Designed scalable automation and test infrastructure for production systems serving 40k+ client endpoints.
- Built CI-integrated tooling with distributed execution, observability, and failure diagnostics.
- Developed end-to-end data pipelines from embedded systems to cloud dashboards.
- Experience delivering software in regulated environments (FDA validation).

WORK EXPERIENCE

FreedomPay: QA Engineer

Jan 2023 - July 2025

- Designed and implemented a reusable Java-based automation framework integrated with CI pipelines, supporting multiple production systems and facilitating the onboarding of 40,000+ client locations.
- Built a reporting feature to collect and publish build/run metadata, improving visibility into test results and system performance across the web department.
- Led evaluation of a third-party customer-facing web application, assessing quality, performance, and integration viability. Delivered technical feedback to stakeholders on readiness for production deployment and advised on the decision of whether to adopt.

PNC: Automation Engineer

April 2022 - December 2022

- Improved delivery velocity (20%) by standardizing team workflows and improving engineering feedback loops through joint demos and reviews.
- Updated automation frameworks across multiple teams, adding database and Kafka connections in addition to user-focused tests, and helped introduce quality control through pull requests and code reviews.

Corista: QA Engineer

Nov 2021 - March 2022

- Offloaded performance testing from local machines onto containerized remote environments with Docker and Kubernetes, improving scalability, reproducibility, and observability.
- Created pytest infrastructure for backend services in dev and prod environments.

Insulet: Test Automation Engineer

June 2020 - Nov 2021

- Significantly increased throughput and standardization by automating manual workflows for team of 15 developers.
- Executed validation and accompanying documentation of automation framework for each release to be submitted to FDA.
- Designed and implemented new features of Python framework to automate environment with various hardware, peripherals, mobile devices, and GUI applications.

PROJECTS

Canine Performance Tracking Mobile App - *In Progress*

Developing a domain-driven, cross-platform mobile application using React Native and TypeScript, with offline-first data modeling, sync reconciliation, and analytics for longitudinal performance tracking.

Plant Data Monitoring Tools

Designed and implemented end-to-end IoT data pipeline using ESP32 (ESP-IDF, FreeRTOS) to collect sensor telemetry, publish secure MQTT messages, ingest data on a server, persist to a database, and visualize metrics on Grafana dashboards for real-time and historical analysis.

Desktop Ecommerce Platform

Built a GUI-based e-commerce platform using Java, SQL, and Docker, enabling customers to browse and place orders while providing admin-level inventory management.

EDUCATION

Computer Science M.S. - West Chester University	2024 - present
– <i>1st Place, Graduate Poster & Presentation - Student Research Day (Fall 2025)</i>	
Computer Science B.A. - Boston College	2016 - 2020

SKILLS

Languages: Java, Python, TypeScript, JavaScript, C, C++, Rust, SQL, Bash
Frontend: React, React Native, Next.js
Testing & Automation: Cypress, Selenium, Cucumber, pytest, Katalon
Infrastructure & Tooling: Azure DevOps, Git, Docker, Kubernetes, Dynatrace, Grafana